

Machine Learning – Classification:

Problem Statement or Requirement:

A requirement from the Hospital, Management asked us to create a predictive model which will predict the Chronic Kidney Disease (CKD) based on the several parameters. The Client has provided the dataset of the same.,”CKD.csv”.

1. **Stage 1:** As the inputs in the datasets are numbers, have chosen Machine Learning Domain.
Stage 2: As the requirements are clear and we have the output in the dataset, supervised learning method is used.
Stage 3: As the output to be predicted is categorical values, it is obvious that we need to create the model using Classification technique/algorithm.
2. The dataset has **399 rows × 25 columns**.
3. Here, in the dataset we have some of the categorical columns, which we have to convert to numerical columns. For this pre-processing of data, we use **get_dummies** function from pandas library to convert them to numerical values, before splitting the input and output columns. Here, as we have nominal values, **One-hot Encoding** technique is used for conversion.

Also, for standardizing the inputs, **standardization** method is used before model creation after train and test split of the dataset, so that the inputs will be in a defined range, easier to train the model.

4. First created model using **Support Vector Machine** Algorithm, where the accuracy is **62%** and the roc_auc_score is **0.5**.
Here, The f1_macro value for best parameter {'C': 10, 'gamma': 'auto', 'kernel': 'rbf'}: 0.4807. The classification_report is shown below:

	precision	recall	f1-score	support
0	0.00	0.00	0.00	45
1	0.62	1.00	0.77	75
accuracy			0.62	120
macro avg	0.31	0.50	0.38	120
weighted avg	0.39	0.62	0.48	120

5. Further trained the model using **Random Forest Classifier** Algorithm, where the accuracy is **97%** and the roc_auc_score is **0.99**. Here, The f1_macro value for best parameter {'criterion': 'log_loss', 'max_features': 'log2', 'n_estimators': 10}: 0.96. The classification_report is shown below:

	precision	recall	f1-score	support
0	0.94	0.98	0.96	45
1	0.99	0.96	0.97	75
accuracy			0.97	120
macro avg	0.96	0.97	0.96	120
weighted avg	0.97	0.97	0.97	120

Next, created model using **Decision Tree** algorithm where the accuracy is **62%** and the roc_auc_score is **0.5**.

Here, the f1_macro value for best parameter {'criterion': 'log_loss', 'splitter': 'random'}: 0.48

The classification_report is shown below:

	precision	recall	f1-score	support
0	0.00	0.00	0.00	45
1	0.62	1.00	0.77	75
accuracy			0.62	120
macro avg	0.31	0.50	0.38	120
weighted avg	0.39	0.62	0.48	120

Further created models using **Logistic Regression, K-Nearest Neighbour, Naive Bayes** algorithm with the parameter as follows:

Logistic Regression:

The accuracy is **99%** and the roc_auc_score is **1.0**. Here, The f1_macro value for best parameter {'penalty': 'l2', 'solver': 'lbfgs'}: 0.99.

The classification_report is shown below:

	precision	recall	f1-score	support
0	0.98	1.00	0.99	45
1	1.00	0.99	0.99	75
accuracy			0.99	120
macro avg	0.99	0.99	0.99	120
weighted avg	0.99	0.99	0.99	120

K-Nearest Neighbour:

The accuracy is **95%** and the roc_auc_score is **0.99**. Here, the f1_macro value for best parameter {'algorithm': 'auto', 'metric': 'minkowski', 'p': 2}: 0.95.

The classification_report is shown below:

	precision	recall	f1-score	support
0	0.88	1.00	0.94	45
1	1.00	0.92	0.96	75
accuracy			0.95	120
macro avg	0.94	0.96	0.95	120
weighted avg	0.96	0.95	0.95	120

Naive Bayes:

Models were created based on GaussianNB, MultinomialNB, BernoulliNB, ComplementNB classifiers, without using any hyper tuning parameters(gridsearchcv).

GaussianNB:

	precision	recall	f1-score	support
0	0.96	1.00	0.98	45
1	1.00	0.97	0.99	75
accuracy			0.98	120
macro avg	0.98	0.99	0.98	120
weighted avg	0.98	0.98	0.98	120

The roc_auc_score is **0.98**.

MultinomialNB:

	precision	recall	f1-score	support
0	0.67	0.98	0.79	45
1	0.98	0.71	0.82	75
accuracy			0.81	120
macro avg	0.82	0.84	0.81	120
weighted avg	0.86	0.81	0.81	120

The roc_auc_score is **0.84**.

BernoulliNB:

	precision	recall	f1-score	support
0	0.85	1.00	0.92	45
1	1.00	0.89	0.94	75
accuracy			0.93	120
macro avg	0.92	0.95	0.93	120
weighted avg	0.94	0.93	0.93	120

The roc_auc_score is **0.94**.

ComplementNB:

	precision	recall	f1-score	support
0	0.67	0.98	0.79	45
1	0.98	0.71	0.82	75
accuracy			0.81	120
macro avg	0.82	0.84	0.81	120
weighted avg	0.86	0.81	0.81	120

The roc_auc_score is **0.84**.

- Overall while comparing the above algorithms Logistic Regression algorithm, created the model with high accuracy score **99%** and higher roc_auc_score, **1.0**. Hence Logistic Regression model created is saved in a file with extension .sav.

The saved model is used by the client/end user to load the model and predict the CKD classification for future use.